

Title: Current challenges for unseen-epitope TCR interaction prediction and a new perspective derived from image classification

Aim

The primary aim of the research presented in the sources is to address the current challenges in predicting how T-cell receptors (TCRs) recognize epitopes, with a specific focus on the difficult task of generalizing predictions to unseen epitopes.

To achieve this, the study focuses on several key objectives:

- **Development of a New Model (ImRex):** The researchers present ImRex (interaction map recognition), a novel feature representation approach inspired by image classification. It uses a convolutional neural network (CNN) to process "interaction maps" formed by the pairwise combination of physicochemical properties of amino acids in TCR and epitope sequences.
- **Establishment of Rigorous Validation Strategies:** The study aims to define and examine appropriate validation methods to accurately measure a model's ability to extrapolate. Specifically, it distinguishes between seen-epitope prediction (how well a model recognizes epitopes it was trained on) and unseen-epitope prediction (how well it recognizes novel epitopes).
- **Identification and Mitigation of Biases:** The research scrutinizes various challenges and biases inherent in TCR-epitope data that can lead to misleadingly optimistic results. These include:
 - The selection and generation of negative data.
 - The impact of imbalanced epitope distributions in datasets.
 - The exchangeability (or lack thereof) between TCR alpha and beta chains.
- **Benchmarking Performance:** The research aims to provide a benchmark for generic TCR-epitope models, comparing the new ImRex approach against prevailing "dual input" models and established epitope-specific classifiers.

Ultimately, the work seeks to provide a new perspective and more stringent framework for developing and validating predictive models that could one day support personalized healthcare, such as tailored cancer immunotherapies or vaccine assembly.

Theoretical Framework / Key Concepts

The theoretical framework of this research is grounded in immunoinformatics and deep learning, specifically applying principles from image classification to the biological problem of T-cell receptor (TCR) and epitope interaction prediction.

Core Biological Framework

- **TCR-pMHC Interaction:** The fundamental process where a TCR on a T-cell recognizes an immunogenic peptide (epitope) presented by a Major Histocompatibility Complex (MHC) molecule.
- **Sequence Diversity:** The framework addresses the extreme diversity of the immune repertoire, driven by somatic V(D)J recombination in TCRs and the rapid evolution of pathogens or malignant cells.
- **Cross-reactivity:** A complicating concept where a single epitope can be bound by varying TCRs, and conversely, a single TCR can recognize multiple epitopes.

Computational and Machine Learning Concepts

- **Seen-Epitope vs. Unseen-Epitope Prediction:** This is a critical distinction in the study's validation framework.
 - **Seen-epitope:** Predicting interactions for epitopes already encountered during training, albeit with different TCR sequences.
 - **Unseen-epitope:** Predicting interactions for entirely novel epitopes, testing the model's ability to extrapolate or generalize.

- Covariate Shift: The phenomenon where a predictive model underperforms on new data because the distribution of inputs (TCRs and epitopes) is divergent from the training set.
- Interaction Map (ImRex): A novel feature representation that treats molecular sequences like an image. It uses the pairwise combination of amino acid physicochemical properties (hydrophobicity, hydrophilicity, mass, and isoelectric point) to represent the binding context directly rather than encoding sequences independently.
- Dual Input Models: The "prevailing approach" where the two interacting sequences are supplied to a neural network as distinct entities before being combined or concatenated later in the process.

Data Validation and Bias Mitigation

- Negative Data Generation: Because databases only contain positive (interacting) pairs, researchers must generate "biologically plausible" negatives.
 - Shuffling: Combining a known TCR with a randomly selected epitope it is not known to bind with.
 - Reference Negatives: Sampling TCRs from healthy repertoires not found in the positive dataset.
- Epitope Imbalance/Skew: The concept that curated datasets are dominated by a few highly abundant epitopes, which can lead to overfitting or biased performance metrics.
- Decoy Epitope Models: A validation strategy where epitopes are replaced by random permutations to determine if a model has captured "true molecular forces" or is simply memorizing TCR motifs.
- Non-exchangeability of Chains: The finding that TCR alpha (TRA) and beta (TRB) chain information cannot be used interchangeably in these models, likely due to differences in their underlying binding processes.

Methodological Concepts

The methodological concepts in the provided source material center on the development, validation, and bias mitigation of machine learning models for predicting T-cell receptor (TCR) and epitope interactions.

Core Feature Representation

- Interaction Map (ImRex): A novel representation that treats molecular sequences as images. By lining up the TCR and epitope sequences on the axes of a 2D matrix, every position represents a pairwise combination of their physicochemical properties.
- Physicochemical Properties: The models use four biochemically distinct properties to characterize amino acids: hydrophobicity, hydrophilicity, mass, and isoelectric point.
- Absolute Difference Operator: This is the specific operator used to combine the properties of two amino acids into the interaction map, helping to highlight regions of compatibility, such as opposite charges.

Validation and Generalisation Strategies

- Seen-epitope vs. Unseen-epitope Prediction: This distinction is critical for evaluating performance. Seen-epitope prediction tests the model on TCRs paired with epitopes it has already encountered in training, while unseen-epitope prediction tests the model's ability to extrapolate to entirely novel epitopes.
- Epitope-grouped Cross-Validation (CV): A stringent validation strategy where folds are created such that TCR-epitope pairs in different folds never share an epitope, ensuring the model is tested on "unseen" targets.
- Covariate Shift: A concept describing when predictive models fail on new data because the distribution of the inputs (TCRs or epitopes) is divergent from the training set.
- Decoy Epitope Models: A validation method where epitopes are replaced by random permutations to see if the model has captured "true molecular forces" or is simply memorizing TCR motifs.

Data Handling and Bias Mitigation

- Negative Data Generation: Since databases only contain positive interactions, researchers must create "biologically plausible" negatives. The sources contrast two methods:
 - Shuffling: Combining a known TCR with a randomly selected epitope it is not known to bind with.
 - Reference Negatives: Sampling TCRs from a healthy repertoire not found in the positive dataset.

- Epitope Imbalance (Skew): The observation that datasets are often dominated by a few highly abundant epitopes, which can bias performance metrics.
- Downsampling: A strategy used to reduce the number of observations for the most frequent epitopes, allowing for more robust and balanced cross-validation.
- Edit Distance: A metric used to measure the similarity between epitopes; the research found that models extrapolate better to unseen epitopes that have a low edit distance (high similarity) to those in the training set.

Modelling Architectures

- Convolutional Neural Network (CNN): The primary deep learning architecture used to extract features from the interaction maps, following a design similar to image classification models.
- Dual Input Architecture: The "prevailing approach" where the two interacting sequences are encoded separately (often using BLOSUM matrices) and supplied to the network as distinct entities before being combined later.
- Non-exchangeability of Chains: The methodological finding that TCR alpha (TRA) and beta (TRB) chain information cannot be used interchangeably, as models trained on one do not generalize to the other.

Assumptions

The research into T-cell receptor (TCR) and epitope interaction prediction relies on several key statistical, biological, and methodological assumptions:

Data and Validation Assumptions

- Negative Data Generation (Shuffling): When generating negative examples by shuffling known positive pairs, the researchers assume that TCRs can only bind to their specific seen epitope partner. Consequently, they assume any other random combination within the dataset would not result in a successful binding event.
- Negative Data Generation (Reference Repertoire): When using external reference repertoires for negative data, it is assumed that TCRs sampled from healthy individuals are unlikely to bind to a randomly assigned epitope from the positive dataset.
- Negligible Overlap Bias: The authors assume that the very small overlap (consistently fewer than 0.15%) between the positive sequences and the negative reference sequences in cross-validation folds results in negligible bias.
- Requirement for Diversity: Based on the finding that downsampling had a negligible effect on performance, the authors assume that future improvements in prediction will depend more on collecting diverse epitope-TCR pairs rather than just increasing the volume of existing data types.

Methodological and Modeling Assumptions

- Superiority of Interaction Maps: The researchers postulate that their interaction map (ImRex) provides a better representation of the binding context than the prevailing "dual input" approach, which learns embeddings for each molecule independently.
- Adequacy of Physicochemical Properties: The model assumes that the "true underlying molecular forces" governing the binding process can be extracted from four specific amino acid properties: hydrophobicity, hydrophilicity, mass, and isoelectric point.
- Pairwise Compatibility: The choice of the absolute difference operator to create the interaction maps is based on the assumption that it highlights regions of molecular compatibility, such as amino acids with opposite charges (isoelectric points).
- Epitope Similarity and Interaction Patterns: The research assumes that epitopes with low edit distance (high similarity) share similar physicochemical properties and, therefore, some underlying interaction patterns may be transferred between them.

Biological Assumptions

- **Chain Contribution:** It is generally assumed in the field that both alpha and beta chains contribute to binding specificity, even though most available data currently only covers the beta chain.
- **Non-exchangeability of Chains:** The study concludes that TCR alpha (TRA) and beta (TRB) chain information is not exchangeable, assuming this is because the underlying binding processes for the two chains differ too significantly for a model trained on one to be transferred to the other.
- **CDR3 as the Primary Information Source:** The models are built on the assumption that the amino acid sequences of the TCR's CDR3 region contain the relevant information necessary to predict epitope recognition.

Methodology

The methodology utilized in the research centres on the development and rigorous evaluation of a new feature representation called ImRex (interaction map recognition), designed to predict T-cell receptor (TCR) and epitope interactions.

Data Collection and Processing

Data was primarily sourced from the August 2019 release of VDJdb, which initially provided 75,474 curated pairs of TCR CDR3 sequences and their targets. This was narrowed to 19,842 unique human MHC class I pairs after filtering for specific sequence lengths (10-20 amino acids for CDR3 and 8-11 for epitopes) and removing duplicates. The final filtered dataset was divided into alpha chain (5,654 sequences) and beta chain (14,188 sequences) subsets. Additionally, an external test set was derived from the McPAS-TCR dataset to evaluate independent generalisation.

Feature Representation (ImRex)

The ImRex approach converts molecular sequences into a 3D tensor called an interaction map, inspired by image classification techniques. This representation is created through the following steps:

- **Property Selection:** Each sequence is converted into vectors based on four physicochemical properties: hydrophobicity, hydrophilicity, mass, and isoelectric point.
- **Pairwise Operator:** A matrix is computed containing the pairwise absolute differences between the vector elements of the TCR and the epitope. This operator was chosen to highlight regions of molecular compatibility, such as opposite charges.
- **Standardisation:** The resulting matrices are scaled between zero and one and zero-padded to a dimension of 20×11 before being combined into a 3D tensor.

Model Architectures

The researchers compared two different Convolutional Neural Network (CNN) architectures:

- **ImRex CNN:** This model accepts the $20 \times 11 \times 4$ interaction maps as input. Its architecture features two identical units containing 2D convolutional layers, 2D max pooling, spatial dropout (25%), and batch normalization. A final fully connected layer with 32 neurons leads to a single sigmoid output neuron.
- **Dual Input CNN:** Used as a benchmark, this model represents the prevailing approach where sequences are provided as separate entities. It utilizes BLOSUM50 encoding and 1D convolutions to extract features from each sequence independently before concatenating them for final processing.

Bias Mitigation and Training

To address the lack of true negative data in databases, the study compared shuffling positive pairs (pairing a known TCR with a random, non-partner epitope) against using reference negatives from a healthy TCR repertoire. Shuffling was ultimately recommended because reference negatives resulted in overly optimistic performance estimates. To handle epitope imbalance, a downsampling strategy was employed to reduce the highest epitope counts, allowing for more robust cross-validation.

Validation Strategies

Performance was assessed using two distinct cross-validation (CV) schemes to distinguish between different types of generalisation:

- Regular repeated 5-fold CV: Measures "seen-epitope" performance, where the model predicts interactions for epitopes it encountered during training.
- Epitope-grouped CV: Measures "unseen-epitope" performance by ensuring that TCR-epitope pairs in the test folds never share an epitope with the training folds. The researchers also employed decoy epitope models, where epitopes were replaced by random permutations to check if the model was capturing true molecular forces or merely memorising TCR motifs. Performance was quantified using the AUROC and average precision (AP).

Main Findings

The research identifies that while predicting T-cell receptor (TCR) recognition for unseen epitopes remains a significant challenge, the novel ImRex (interaction map recognition) approach provides a better framework for extrapolation than previous methods.

The main findings from the sources are:

1. Generalisability and Performance

- Difficulty of Unseen-Epitope Prediction: Extrapolating to novel epitopes not encountered during training is far more difficult than predicting interactions for seen epitopes. Performance for unseen-epitope prediction is generally poor across models, reflecting the "immense space" of the immune repertoire.
- ImRex Feasibility: The ImRex model makes extrapolation feasible for a subset of epitopes that are physically similar (low edit distance) to those in the training set.
- Comparison to Dual Input: The ImRex approach, which uses pairwise physicochemical property maps, outperforms the prevailing "dual input" models (which encode sequences separately) in both seen and unseen epitope settings.
- Epitope-Specific Superiority: Despite advancements in generic models, epitope-specific models (like TCRex) still outperform generic ones because they solve a simpler, less diverse problem.

2. Methodological Biases and Data Handling

- Negative Data Selection: The study found that using an external reference repertoire for negative data leads to over-optimistic performance estimates. Generating negatives by shuffling known positive pairs is recommended to prevent models from simply memorising TCR motifs.
- Epitope Imbalance: Curated datasets are highly skewed, with a few epitopes appearing thousands of times. This skew can cause models to overfit on abundant epitopes and makes overall performance metrics misleading.
- Non-Exchangeability of Chains: TCR alpha (TRA) and beta (TRB) chains are not exchangeable. Models trained on beta chains do not generalise to alpha chains, likely because the underlying binding processes differ too significantly.

3. Evaluation Standards

- Per-Epitope Metrics: Researchers found that overall AUROC/AP metrics are insufficient because they can be skewed by a single abundant epitope. A per-epitope ranking is necessary to understand a model's true behaviour.
- Decoy Models: The use of decoy epitope data (permuting epitope sequences) is essential for establishing a baseline; it reveals whether a model has captured "true molecular forces" or is merely memorising TCR motifs.

- **Downsampling Effects:** Downsampling to address epitope skew has a negligible effect on performance, suggesting that future improvements rely more on collecting diverse epitope-TCR pairs rather than larger volumes of existing data.

Limitations

According to the sources, the prediction of T-cell receptor (TCR) and epitope interactions faces several significant limitations, ranging from the quality and quantity of available data to the inherent biological complexity of the immune system.

The primary limitations highlighted in the sources include:

1. Data Scarcity and Diversity

- **Limited Training Data:** Compared to MHC-epitope data, there is a lack of high-quality training data for TCR-epitope recognition. Data generation for TCRs occurs at a much smaller scale, which limits the ability of complex models to learn underlying patterns.
- **High Sequence Diversity:** The "quasi-random somatic V(D)J recombination" of TCRs and the rapid evolution of pathogens create an immense sequence space. The high diversity and non-homogeneous nature of these sequences make it difficult for models to extract distinguishing features.

2. Generalization to Unseen Epitopes

- **Extrapolation Failure:** While models can achieve reasonable performance on epitopes already encountered during training ("seen epitopes"), they generally cannot yet reliably extrapolate to novel, "unseen" epitopes.
- **Performance Disparity:** The performance on unseen epitopes is significantly poorer than on seen epitopes. The sources suggest this is likely due to "covariate shift," where models under-perform because the distribution of input TCRs or epitopes in the test set differs from the training set.
- **Similarity Dependency:** Even advanced models like ImRex perform best only on epitopes that are very similar to the training data (e.g., within a single amino acid difference); performance drops as the edit distance increases.

3. Dataset Biases and Technical Hurdles

- **Imbalanced Distribution:** Curated datasets are highly skewed, with a few "abundant" epitopes appearing in thousands of entries while the majority appear in only a few. This can cause models to overfit on common epitopes.
- **Absence of True Negatives:** Current datasets only contain positive interacting pairs. Researchers must generate "negative" data (non-interacting pairs) through methods like shuffling or reference repertoires, which can introduce over-optimistic performance estimates or unknown biases.
- **Memorization vs. Learning:** There is a danger that models memorize specific CDR3 motifs regardless of the epitope partner, rather than learning the true molecular forces governing binding.

4. Biological and Structural Limitations

- **Single-Chain Data Constraint:** Most available data consists of single-chain (usually beta-chain) sequences, even though both alpha and beta chains contribute to binding specificity in reality.
- **Chain Non-Exchangeability:** Models trained on beta-chain data cannot be transferred to alpha-chain data, and adding alpha-chain sequences as training examples does not markedly improve model performance, indicating that the underlying binding processes may differ significantly between the two.
- **Complexity of Recognition:** TCR-epitope recognition is fundamentally more complex than MHC-epitope recognition due to the larger diversity in sequence pairing and the cross-reactivity of TCRs.

5. Evaluation Pitfalls

- **Misleading Metrics:** Overall performance metrics (like AUROC) can provide a distorted picture because they are heavily influenced by the performance of a few abundant epitopes. A model might appear accurate overall but fail completely on the majority of individual epitopes.
- **Decoy Performance:** Some models achieve high performance even when trained on "decoy" (nonsense) epitopes, suggesting they are relying on TCR patterns alone rather than true interaction patterns.

Implication of Practise

Advanced Feature Representation

Practitioners are advised to move away from modelling sequences independently and instead adopt interaction-based feature representations, such as the Interaction Map (ImRex) framework.

- **Pairwise Relationships:** These representations capture the physicochemical relationships between amino acids directly.
- **Spatial Patterns:** By converting sequences into structured matrices, convolutional neural networks can extract spatial interaction patterns that reflect the binding context.

Rigorous Validation Standards

Standard random cross-validation is often insufficient as it can hide information leakage and produce over-optimistic results.

- **Structured Cross-Validation:** Practitioners should use grouped cross-validation (e.g., splitting by epitope, patient, or time) to ensure the test data contains entirely new targets.
- **Testing for Extrapolation:** Models must be explicitly tested on truly unseen targets that differ significantly from training data to simulate real-world deployment.
- **Diagnostic Testing:** To ensure a model is learning genuine patterns rather than memorising artefacts, practitioners should use decoy testing (e.g., randomising input features or permuting labels) as a routine diagnostic.

Strategic Data Engineering

The quality and structure of data are more critical for generalisation than the sheer quantity of observations.

- **Negative Data Design:** Generating negative examples is a core modelling decision. Practitioners must avoid "artificially easy" negatives and use realistic pairing strategies to prevent bias.
- **Diversity Over Size:** Increasing data for well-represented categories provides diminishing returns. Instead, data collection should prioritise diversity, including rare cases and edge scenarios to expand coverage of the input space.
- **Component Separation:** Distinct system components (such as TCR alpha and beta chains) should be treated as separate modelling problems and validated independently rather than being pooled together.

Operational Reliability and Reporting

To maintain safety and transparency in deployment, models should include mechanisms for self-assessment.

- **Similarity-Based Triage:** Practitioners should implement similarity metrics to compare new inputs to training data. Predictions for high-similarity inputs are usable with confidence, while low-similarity cases should be flagged as exploratory.
- **Granular Performance Metrics:** Overall aggregate scores can be misleading. Reporting should focus on per-category, per-target, or worst-case performance to reveal where a model might fail on less common but important cases.
- **Specialised vs. General Models:** Specialised models should be reserved for high-stakes, well-defined tasks where high accuracy is required, while general-purpose models are better suited for exploratory screening.